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Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738068928979-Chapter 4 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

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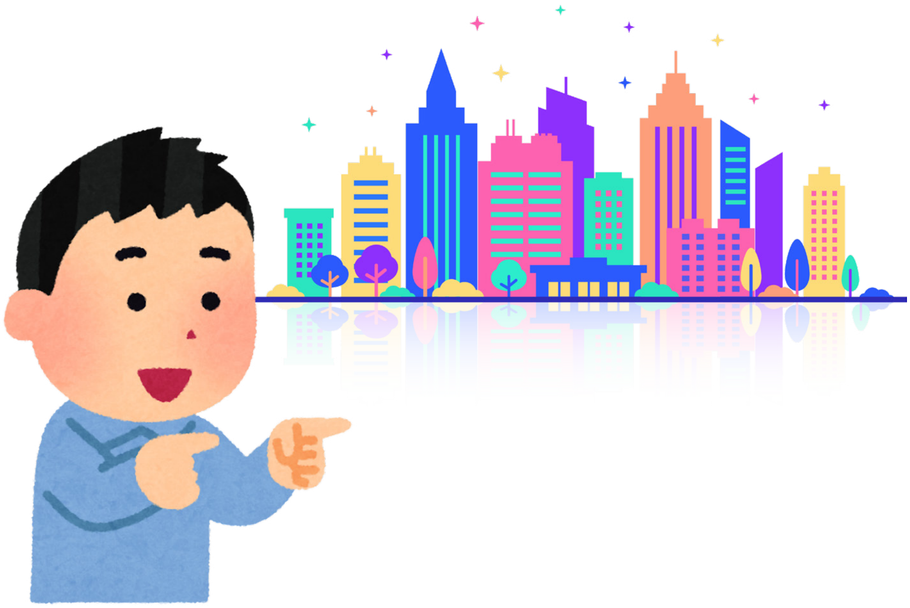
Project Introduction

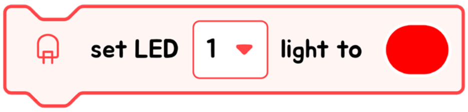
LIGHTING UP THE CITY: HUGH'S COLORFUL DISCOVERY

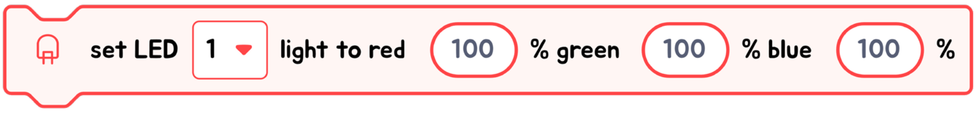
One evening, as Hugh was setting up for his school's science fair, he encountered a problem that seemed impossible at first. He had created a small model of a cityscape, complete with miniature buildings and streetlights, and he wanted the lights to change colors to demonstrate how different lighting could affect the mood and appearance of the city. But no matter what he tried, the colors from his simple bulbs were dull and didn't blend the way he envisioned. It was crucial for his project to work by the fair the next morning, yet here he was, stuck and unsure what to do next.



Just then, his science teacher, Mr. Benson, walked by and noticed Hugh's frustration. Seeing a teachable moment, Mr. Benson introduced Hugh to the concept of RGB LED lights, explaining how combining red, green, and blue could achieve the vibrant, dynamic colors Hugh needed for his model. Inspired by the idea, Hugh spent the night adjusting his project, using RGB LEDs to illuminate his model city in a spectrum of beautifully blended colors. The next day, not only did his project win praise at the fair, but Hugh also realized the power of persistence and the impact of using science to bring creativity to life. His experience perfectly illustrated the lesson in their upcoming chapter on color and light—a lesson Hugh was now especially excited to explore further.

ABOUT LED MODULE



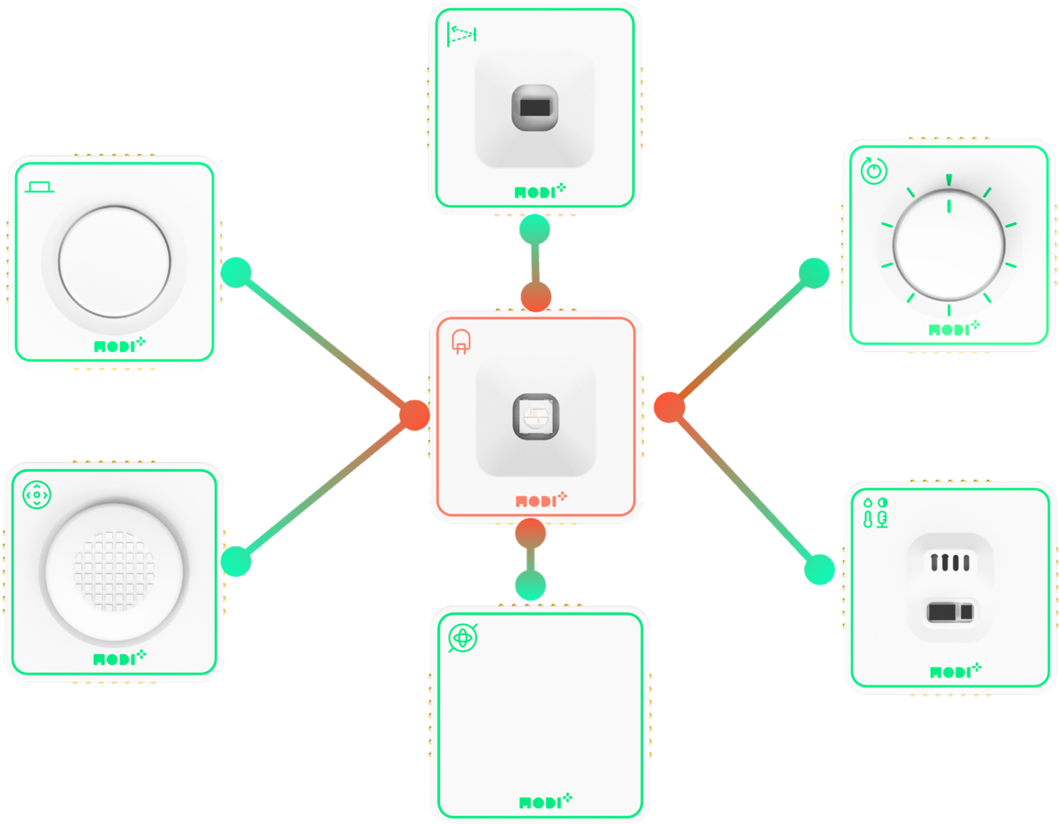


The LED module features three distinct blocks in the block area. The first block enables you to select a color using hue, saturation, and brightness values. The second block allows you to set the color based on RGB (Red, Green, Blue) values. The third block is used to turn off the LED module

INTERACTIONS WITH OTHER MODULES

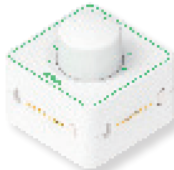
As an output module, the LED can interact with all input modules. You can observe these interactions by connecting the LED module to any input module, even without any programming required!

INTERACTION BETWEEN LED AND OTHER MODULES

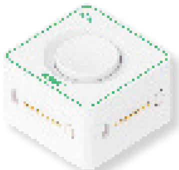


Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?

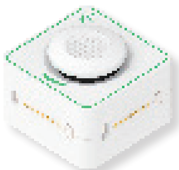


Dial

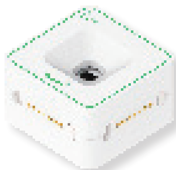


Button

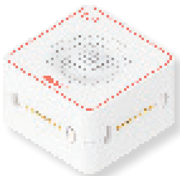
Environment



Joystick



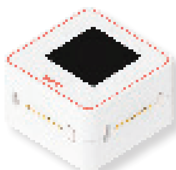
ToF



Speaker

IMU

LED

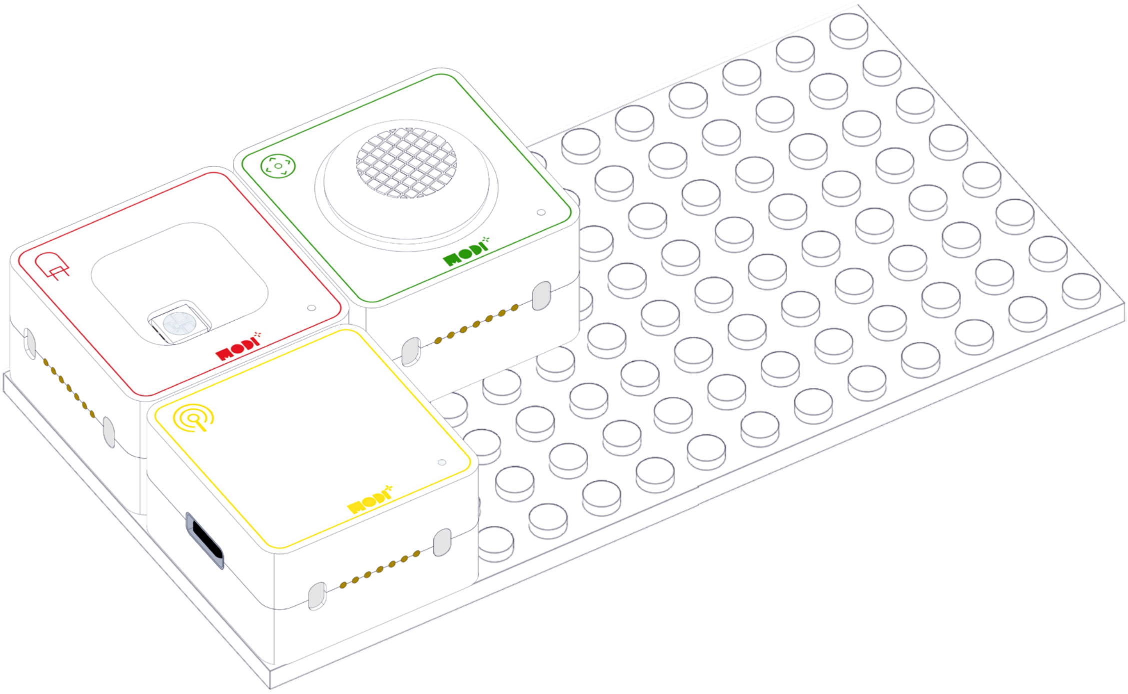


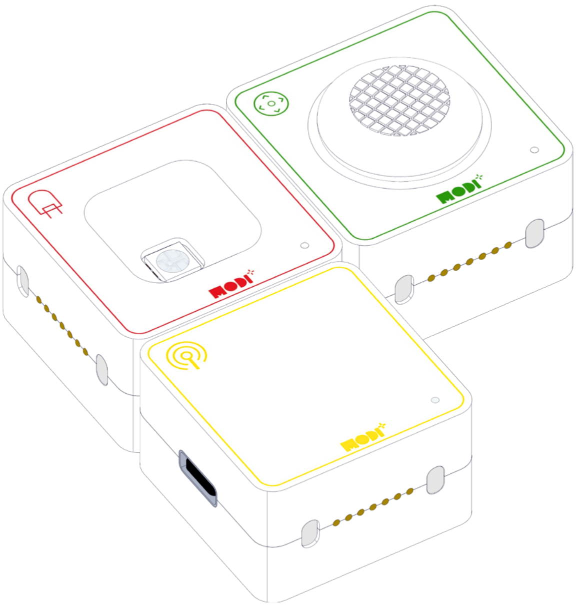
Display

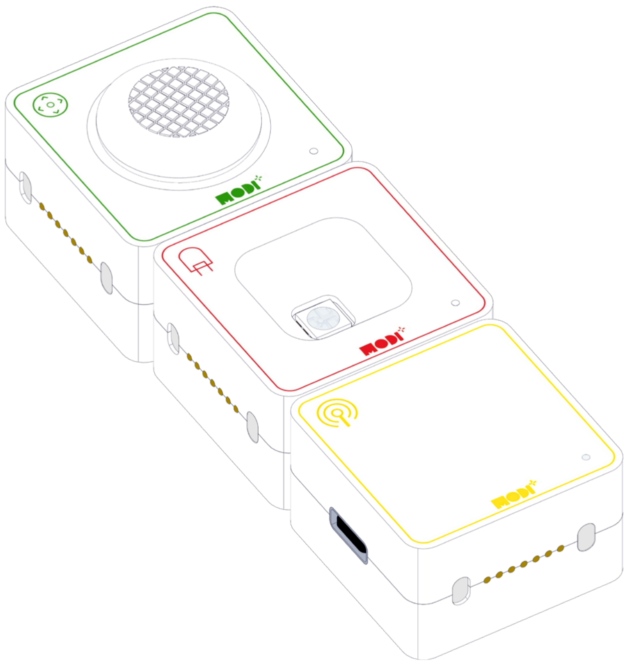
Motor A

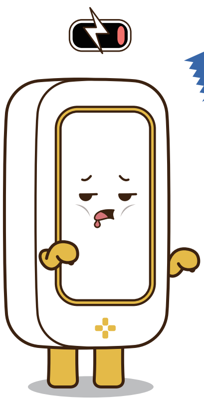
Motor B

HOW DOES THE CREATION LOOK LIKE?







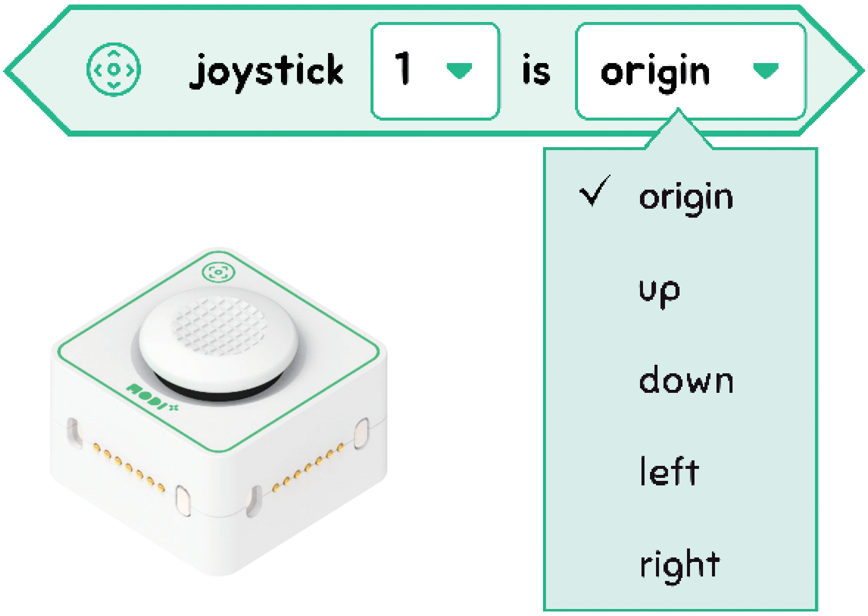


TIPS

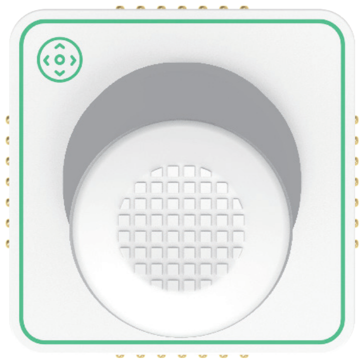
The order of connections between the modules does not matter. As long as all the required modules are physically connected, they will function the same, regardless of which one is connected first.

Module Functionality

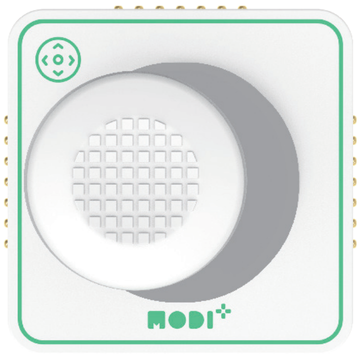
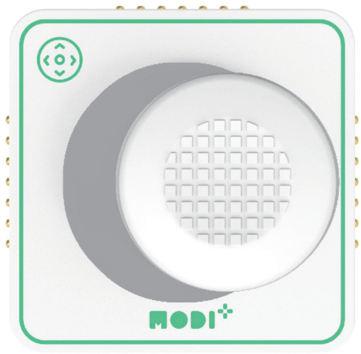
Let's look into the details of the Joystick module.



Including the neutral position 'origin,' the joystick has 5 different positions.

Upward	Downward
	

Pushing it upward and downward affects movement on the **y-axis**
- Range : -100 % (fully downward) ~ +100 % (fully upward)

Left	Right
	

Moving the joystick left and right controls movement along the **x-axis**
- Range : -100 % (fully left) ~ +100 % (fully right)